

- **Alternative product recommendations:** You've probably tried to buy something you wanted, only to find it wasn't available. But it's unlikely you left the website after that. Instead, you may come across something similar, which you may or may not purchase. Companies utilise a recommendation system to keep customers online by providing them with alternatives to the product they want.
- **Related product recommendations:** Another important application of recommender systems is related product recommendations. When customers make a purchase, it's only natural to offer them something complementary. If a customer purchases a wrist watch, the system may suggest a wallet, which goes well with it.
- **Real-time true personalisation:** With every second of customer activity on e-commerce sites, recommender systems can provide product or content or lifestyle recommendations. They can serve as a shopping assistant, using all the information the customer gives to offer improved recommendations, thus creating a better shopping experience.

Recommender systems: A broad categorisation

Content-based filters

Content-based recommender systems give recommendations based on objects or user metadata. The user's previous purchases are scrutinised. For example, if a user has previously read a book by a particular author or purchased a product from a particular brand, it is considered that the user has a preference for that author or brand and is likely to purchase a similar product in the future. For example, if a user likes the book 'The Adventures of Feluda' by Satyajit Ray, then the system recommends more books by the same author or other

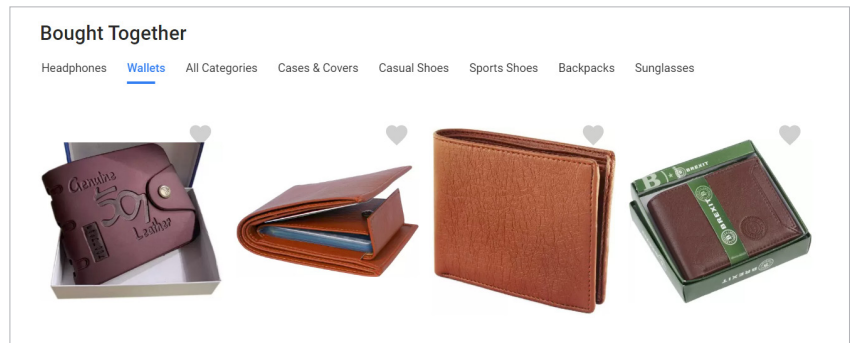


Figure 2: Related product recommendations

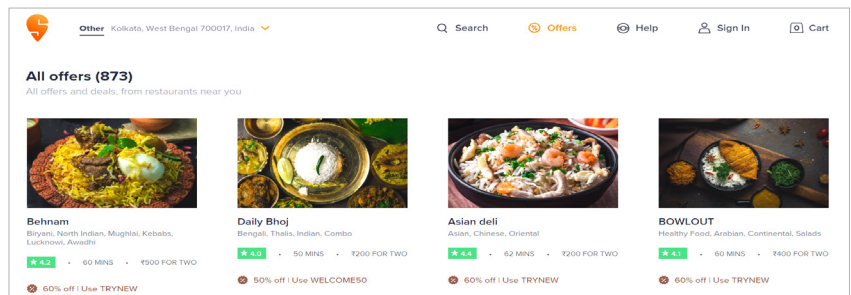


Figure 3: Location specific recommendations

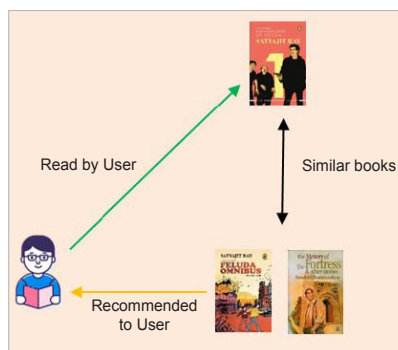


Figure 4: Content based filtering

detective books like 'The Mystery of the Fortress & Other Stories'.

Two types of data are used in this filtering. The first includes the user's preferences, interests, and personal information such as age or, in some cases, the user's history. The user vector represents this information. The second is an item vector, which is a collection of information about a product. The item vector contains all the information that can be used to calculate the similarity between things. Cosine similarity is used to determine the recommendations.

Collaborative filtering

This is regarded as one of the most intelligent recommender systems, and is based on the similarity between different users and things such as e-commerce websites and online movie websites. It makes recommendations based on the preferences of comparable users. Similarity is not limited to the user's preference; it can also be considered between distinct objects. If we have a big amount of knowledge about people and things, the algorithm will make more efficient recommendations.

Collaborative filtering is a method of recommending items that a user might enjoy based on the reactions of other users. It works by sifting through a huge group of people to locate a smaller group of users with likes similar to a specific user. It analyses their favourite things and creates a ranked list of recommendations.

A collaborative filtering system's workflow is usually as follows:

- A user expresses his or her choices by rating system objects (for example, music, books, or movies).